AMARA - AI Driven Virtual Companion for Mental Health Support

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***Abstract*—Newer Methods in the core domains of Generative AI have enhanced the capabilities of the LLMS. Our Large language model is suffering from hallucination. The AI models will give wrong information instead of actual information because they develop from generic training data. Here, in evaluating the study of retrieval augment generation (RAG) the researchers are developing attention based feedback mechanisms in the form of chatbots for mental health care oriented using the Llama. In this paper, we will identify the integration of the mental health care data in the form of csv file and pdf file into the RAG model and the model will make sure that it continues to provide the correct information as well as be aware of context based responses. Additionally, to increase the performance of the RAG model, RAG fusion has been implemented and it will help the model by joining multiple obtained contexts to achieve coherence and relevance. The use of chatbot effectiveness has been tested using hypothesis test analysis of variance and also for the qualitative analysis, the model provides insights to its accuracy as well as reliability. Through this process RAG model will make sure that it will not hallucinate and it has to provide accurate, domain based information for the real conversational aid applications.**

***Index Terms*—RAG, LLM, Hypothesis, chatbot, Generative AI**

I. Introduction

A lot millions of people and mostly young people are suffering from the problems which are related to mental health care such as, anxiety and stress. According to the World Health Organization, which says 264 million people have depression, often with devastating outcomes such as suicide. Mostly the teenage people are suffering from this mental health problems due to the stress and workload and they did not have the proper resources to overcome this problem or the mental health professionals. When you see the middle and low income countries 80% of the people are suffering from these mental health problems and in these countries they don't have proper access to these issues. The increasing disparity of access to mental health services creates a vacuum for scalable solutions such as chatbots, providing real time, accessible mental health support offered through network connectivity. These kinds of chatbots or tools will give more freedom to the people and it will also not judge the people and as an individual they will get more help without waiting or delay. The clash of chatbots over the gap of resource and mental health practitioners focus on critical cases and support broader populations.

Data mostly undergoes training of the self supervised or semi supervised where it has the capability of finding the most complex relationships between the words and it will also identify the phrases. Machine learning algorithms will undergo certain dimensions of the vector search but unlikely the Large language models will undergo the multi dimensional word embeddings and it allows them to use to find the complex relationships between the words and LLMS have the ability to capture the contextual nuances and it will generate precise information as outputs. LLMs have certain limitations: the models will work on the preexisting and mainly generate biases that will raise more concerns about reliability and fairness. Mostly the LLMS will undergo the hallucination where the model will give the false information. This was the major issue of the LLMs and it has occurred because of the domain specific data and moreover there will be irrelevant biases in the training datasets and one more issue with the integration of the real word knowledge large language model will face the issues.

Natural Language Processing and Large Language Models (NLP and LLM) have seen a major advancement with Generative Artificial Intelligence (GenAI), which creates the ability for systems to imitate human conversation and produce derived ideas and content within a second. In contrast to traditional AI models, GenAI takes in diverse range of inputs of text, images, sounds and animations, and transforms them into sophisticated outputs. Foundation Models, which our research discusses at the core of GenAI, are trained on mountains of unlabelled data to be able to do many things. Traditional machine learning and deep learning models work great for simpler outputs, but GenAI excels at dealing with complex inputs and returns a complex result. Retrieval augmented generation

(RAG,) an augmentation of the GenAI model, can integrate additional information external to the model's original training data, making GenAI especially powerful for domain specific applications.

In this study we also utilize RAG to create a healthcare focused chatbot that responds to specialized medical queries in a space which has been lackluster in terms of virtual assistant solutions. We augment the Llama 3.2 Foundation model with the domain specific healthcare data extracted from PDF files and also from the CSV file, building the chatbot. We process these files with the Hugging Face Embeddings to add some vector embedding, store and manage them in the FAISS library. Using Streamlit, we developed a user-friendly interface for the chatbot that lets users easily interact with the chatbot to get their health information. Results of evaluation show that the augmented Llama 3.2 model outperforms other baseline models by showing that the domain specific data works effectively. The approach we advocate for this, is the potential of augmented GenAI to improve its accuracy and accessibility of healthcare knowledge.

II. Review of Literature

A number of works are emerged and affirmed along with the following issues on the application of chatbots in mental health companion.

Koulouri stuides will say about the role of chatbots in mental health support. In their systematic review, Koulouri et al 2022 discussed the possibility of young adults as information-motivation-intention models of stigma and resource demands and the practicality of chatbots within mental health care.

Biswas has designed a CHATGPT based medical assistant chatbot in this study to address the symptoms which include stress, health, education, and depression. Besides, he discussed about the usage of the fine tuned large language models utilizing the open question and answer models for the evaluation of the model and optimization of the heath care outcomes through human centric metrics and automation, to support the implication of AI in health care.

Domains like diagnostic, treatment and public health applications in Mental health have been focused on machine learning. Their study underlined the need for ethical controls in implementation of knowledge management strategy. Organization to improve the complex cognition in interdisciplinary collaboration AI in mental health care.

The technology in mental health was discussed by Graham et al. (2019): diagnosis, prediction, and personalized interventions. As it was noted in their work, they focused on human supervisory responsibility for AI assisted solutions while raising other key points like data security and algorithmic discrimination.

III. Objectives of the Study

The goal of this work is to build a domain specific healthcare chatbot for improved responses to medical queries using Retrieval Augmented Generation (RAG). The study, however, attempts to address limitations of traditional chatbot models like hallucinations, no domain specific knowledge by augmenting the Llama 3.2 Foundation Model with healthcare data derived from PDF documents. The goal of the work is to show that external, applicable data to a foundational model can be integrated to benefit from the performance of the model, providing a portable and user friendly tool that can deliver healthcare information to users.

IV. Data Collection

Dataset 1: Reddit Mental Health Data

The dataset has 4 attributes and 5957 rows. It has information about shared experiences of users and mental disorders,data categorizes mental health topics into five target labels: Stress, Depression, Bipolar, Personality, and Anxiety Issues.

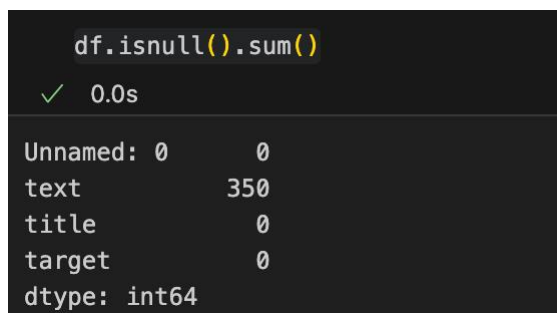
Dataset Link1: https://www.kaggle.com/datasets/neelghoshal/reddit-mental-health-data

Dataset 2: Complete Adult Psychotherapy Treatment Planner

Dataset Link2: https://passthencmhce.com/wp-content/uploads/2022/01/Arthur_E_Jongsma_Jr_L_Mark_Peterson_Tim-1.pdf

V. Explore Data Analysis and Hypothesis test

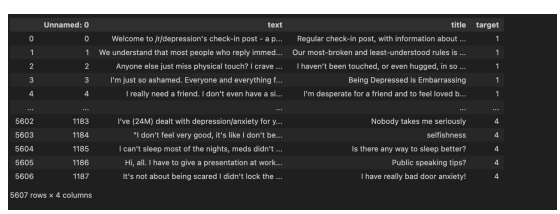
Cleaning and preprocessing the data is crucial before building a big language model or Retrieval-Augmented Generation (RAG), particularly when working with CSV files. Null values must be found and dealt with as the initial step. It is necessary to eliminate missing values and confirm the dataset's integrity. To maintain consistency, it's also critical to standardize certain columns, including text and titles. The below images are about checking for and after the null values.



```
df.isnull().sum()
✓ 0.0s

Unnamed: 0      0
text          350
title           0
target          0
dtype: int64
```

Fig. 1. Checking for the null values in the data



	Unnamed: 0	text	title	target
0	0	Welcome to /r/depression's check-in post - a p...	Regular check-in post, with information about ...	1
1	1	We understand that most people who reply immed...	Our most-broken and least-understood rules is ...	1
2	2	Anyone else just miss physical touch? I crave ...	I haven't been touched, or even hugged, in so ...	1
3	3	I'm just so ashamed. Everyone and everything f...	Being Depressed is Embarrassing	1
4	4	I really need a friend. I don't even have a si...	I'm desperate for a friend and to feel loved b...	1
...	...	...	...	...
5602	1183	I've (24M) dealt with depression/anxiety for y...	Nobody takes me seriously	4
5603	1184	"I don't feel very good, it's like I don't be...	selfishness	4
5604	1185	I can't sleep most of the nights, meds didn't ...	Is there any way to sleep better?	4
5605	1186	Hi, all. I have to give a presentation at work...	Public speaking tips?	4
5606	1187	It's not about being scared I didn't lock the ...	I have really bad door anxiety!	4

5607 rows × 4 columns

Fig. 2. Data after dropping the null values

Data preparation is a crucial next step after handling null values. This entails removing extraneous components including URLs, hyperlinks, physical addresses, email addresses, numbers, stopwords, emoticons, special characters, undue whitespace, and more. This procedure makes sure there is no noise in the input text, which keeps it from impairing model performance.

After that, non-essential keywords like stopwords are eliminated to further cut down noise, and the text is changed to lowercase for consistency. Lemmatization is used to reduce words to their most basic root forms while maintaining contextual meaning, after that tokenization is used to split the cleaned text into smaller tokens. Non-alphabetic characters, punctuation, and question marks are also eliminated.

By ensuring that the language is clear, consistent, and prepared for model input, these procedures facilitate efficient work with big datasets.

Given below shows the features and the frequency

Fig. 3. Visualization for the target feature

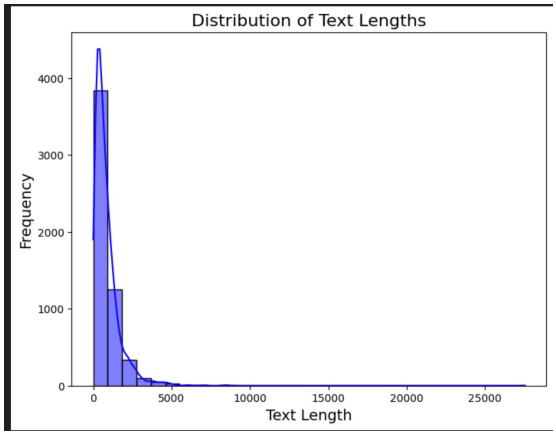


Fig. 4. Visualization for the text feature

A. Hypothesis test

Hypothesis testing is an attempt to establish a procedure through which one can decide whether the available sample

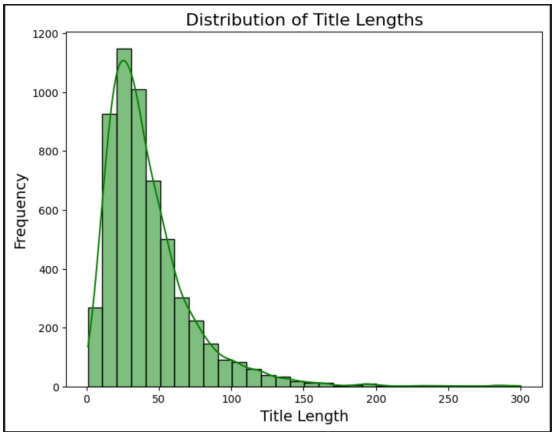


Fig. 5. Visualization for the title feature

data provides sufficient evidence to reject a null hypothesis in consideration to test hypothesis on a given set of assumptions known as the population parameters. In this regard the null hypothesis (H0) was that there is no difference in text lengths between any two categories while the alternative hypothesis was that there is a difference.

Two sample hypothesis testing for the counts of texts of different lengths is achieved with the help of One-Way ANOVA to estimate intergroup differences. If the obtained p-value in the ANOVA test is generally lower than 0.05, the researcher rejects the H0, which means that the text lengths in the groups are different. One also performs a series of independent T-tests** that compare each category against the others and with p-values represented in the heatmap. Data distribution can be better comprehended by using forms of data visualization including boxplots, histograms, and violin plots.

With these tests, we gain knowledge of data structure by figuring out whether or not specific categories possess different text lengths. However, the strength and comparability of the data for different categories and the sizes of the samples are, in fact, important. Some pre-processing errors or variations could affect the results and should be thoroughly controlled.

VI. Chains

Chains are mainly designed to define the workflows of the natural language processing tasks—the NLP tasks include the response generation, Retrieval of data, and everyday tasks such as Q/A. Researchers and developers are currently using frameworks such as langchain, which will allow them to do complex tasks and modular chains to streamline the workflow. It will mainly help the integration of large language models and retrieval augmented generation.

In this project, we use the ConversationalRetrievalChain, which integrates the Ollama model with the retriever and memory and ensures conversational interactions are used. This chain is responsible for the retrieved context and also the ongoing history.

VII. Conversational Buffer Memory

The memory concept allows the model to maintain the chat history between the user and the Large language model or chatbot interactions. It mainly allows the bot to maintain contextual awareness across multiple questions or turns. In this conversation buffer memory, we are using Memory persistence. This will save the context of dialogue for every and multiple response, and the return messages will return the contents of the memory to the chains. The primary use of the buffer memory concept is to give contextual answers based on the priority scenario and the responses.

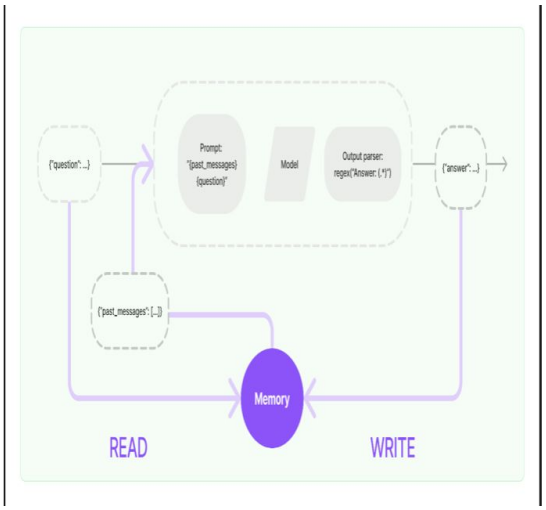


Fig. 6. Memory

Prompt

It will give the text instruction to large language model to generate and guide the model to respond correctly.

VIII. Retrieval-Augmented Generation (RAG)

Retrieval-Augmented Generation (RAG) is a powerful technique that combines information retrieval with text generation. It leverages external knowledge sources to enhance the performance of language models in generating responses based on the retrieved information. This approach is particularly useful in scenarios where a language model needs to answer queries based on a large corpus of documents or a knowledge base.

A. Retrieval

Working of the retrieval is like to get the information from the database response. Whenever the user has given a query to the model, the retrieval component is searching on the vector database and retrieving the relevant documents according to the model and the model ensures it has the necessary data to create the accurate response and retrieves the documents based on semantic search.

B. Augmented

The process of augmented is to enrich the users query by incorporating the retrieved documents. Both the original query and retrieved documents will pass through generative model as an input, allowing it to give the coherent and context relevant responses.

C. Generation

This system generates responses by integrating the user's query with information retrieved from relevant documents. Using advanced language models (LLMs), it refines the retrieved content to ensure the response aligns with the user's input and remains contextually accurate. The model is designed to process the query and the supporting documents cohesively, producing answers faithful to the source material and the user's intent.

D. Vector Storage

Models like `sentence-transformers all-MiniLM-L6-v2`. These vectors will be stored in a high dimensional space and each sentence of the document will be stored as vectors in the FAISS database.

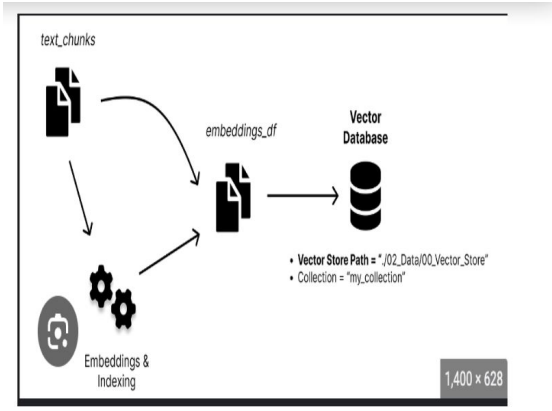


Fig. 7. Vector Storage Database

E. Vector Indexing

Indexes in vector databases are called vector indexing, and these are essentially data structures designed to store and retrieve high-dimensional vectors efficiently. These indexes help organize and manage vector embeddings, enabling fast and scalable similarity. These are similar to rapid searches that are crucial for applications like RAG. They are adding indexes on top of the embedding vectors. It will find similar kinds of vectors, and it will give indexes.

F. Embeddings

In this project, we work with the `Hugging Face Sentence-Transformers` model, trained to generate embeddings that make vectors similar to text.

G. Workflow of RAG

Input: The user provides a query.

1) **Step 1 - Retrieval:** The user's query is embedded into a vector using the same embedding model which is used for storing document vectors. The vector store (e.g., FAISS) performs a similarity search to find the most relevant documents based on the query vector.
2) **Step 2 - Augmentation:** The generative language model is given both the query and the retrieved documents. This step enriches the query by providing relevant context for the task at hand.
3) **Step 3 - Generation:** The large language model generates a response based on the user query and the contextual features from the retrieved documents. The response is designed to be contextually relevant and accurate based on the user's query.

Output: The model generates and provides a response to the user's query, incorporating relevant information from the retrieved documents.

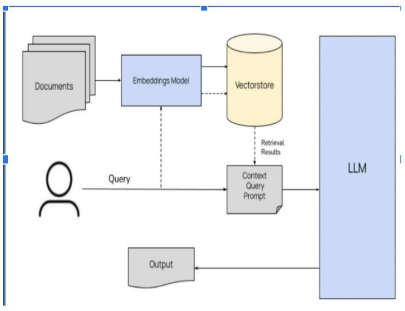


Fig. 8. Rag Architecture

H. Rag fusion

RAG Fusion helps Retrieval Augmented Generation (RAG), which uses multiple retrieval outcomes to enhance coherence and precision in response. Fusion is essentially about rewriting queries and then filtering and ranking things better. RAG fusion helps to close the gap between explicit and implicit search queries. The Reciprocal Rank Fusion (RRF) method combines multiple ranked search results into a single unified list, ensuring that the most relevant items from all sources are prioritized effectively. RRF starts with multiple ranked lists of results from different strategies for each document, list RRF assigns a score based on its rank position and scores for each document are summed across all input lists and documents are stored on basis of combined scores to create the final fused ranking: the higher the rank the greater the weight assigned to source documents during RRF calculation.

I. Voice Interaction for Enhanced User Experience

Stores user input through voice and translates the output generated by assistant into natural human voice. It encompasses the freedom for the user to use voice or text input and improves the user interaction model by incorporating both voice and text paradigms for interaction. We have incorporated Google Text-to-Speech for high quality natural voice output in this model.

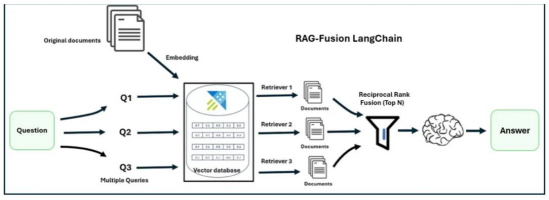


Fig. 9. Vector Storage Database

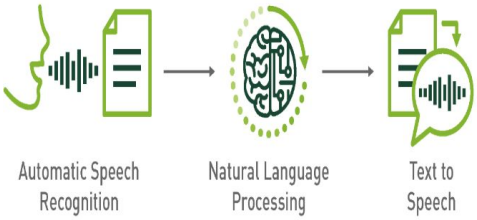


Fig. 10. Voice based interaction

IX. Methodology

In implementing an end-to-end solution for conceiving an interactive mental health assistant chatbot – AMARA – this script invokes Streamlit for the UI interface, LangChain for conversational AI, and other supporting libraries to get voice inputs preprocessed. Text as well as voice query can be entered into the system and the input is given a strong pipeline throughout the system. Data pre-processing include cleaning the words, tokenizing and constructing the word embedding to improve the quality of data retrieval. Social media text data is pre-processed by removing special characters, emojis, URLs, numeric values, and stopwords to convert all inputs to lower case and lemmatizing them.

Tokenization divides text into semantic pieces, where the Hugging Face embedding model generates embeddings for the tokenized text which are indexed in the FAISS vector store for fast search and retrieval. It uses a Retrieval-Augmented Generation thereby the chatbot responds based on rehearsal of previous conversations and an ability to generate new responses. The RAG Fusion technique is used to merge the data obtained from various sources, making the answers complete and correct. Moreover, the structure of the empathetic response option is based on LangChain and Ollama's Llama 3.2 Conversational Retrieval Chain.

The RecursiveCharacterTextSplitter is used to chunk text for processing large documents such as medical books in PDF format. These documents go through pre-processing to obtain their real content and then passes through the embedding

and indexing stage for the documents to be searched. The graphical user interface has been developed using Streamlit, which enables end-users to interact with the chatbot without barriers. The system is also capable of having voice-based interfaces for text to speech as well as for speech to text for people with a wide range of disabilities. A memory buffer is incorporated to preserve conversational context so as to respond with more coherent and interactive answers.

Response handling and debug options guarantee the functionality of the chatbot. The system returns specific answers to queries with source documents as references; therefore, making it transparent and accurate. The latest algorithms, as well as the complicated preprocessing of questions and answers, and the interface make AMARA an effective assistant in mental health.

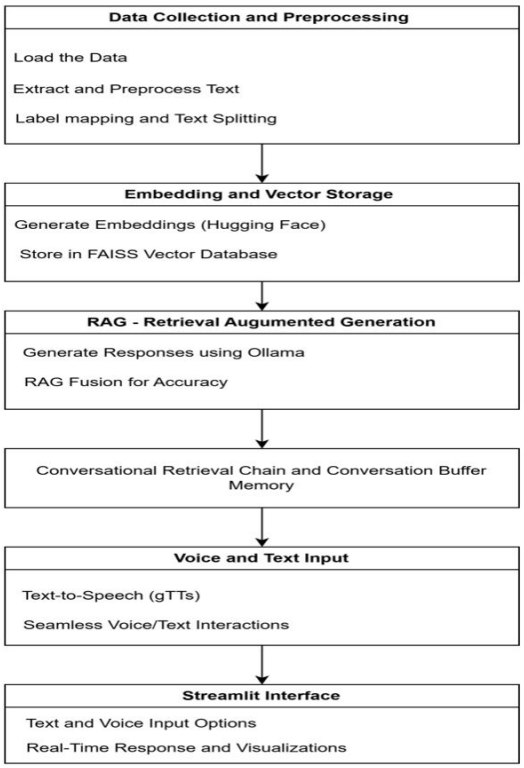


Fig. 11. Voice based interaction

X. Data Visualization and Results

This entails removing extraneous components including URLs, hyperlinks, physical addresses, email addresses, numbers, stopwords, emotions, special characters, whitespace, and more. This procedure makes sure there is no noise in the input text, which keeps it from impairing model performance. It helps in normalizing the text so that certain operations such as analysis or modeling can be accomplished easily.

Length Distribution Analysis: Change in text length before and after preprocessing is done by determining and plotting the frequency distribution of the text. This helps in visualizing how text size has been reduced from cleaning and demonstrates the impacts of the preprocessing steps.

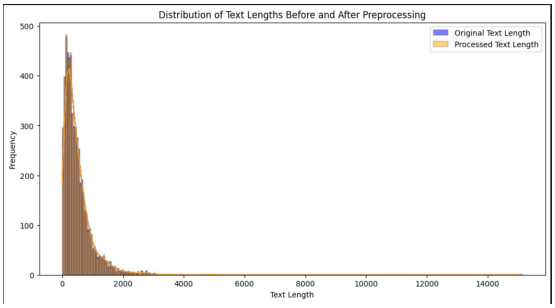


Fig. 12. After text pre-processing of data

Word Cloud Visualization: Following text cleaning, the word occurrences are compiled into a WordCloud. After data cleaning, this estimates the frequency of a particular word or topic occurred in the dataset.

Fig. 13. Most frequent words after text pre-processing

Word Frequency Analysis: After that, the cleaned text is analyzed, and the most frequently used words are described in the form of a bar chart. In this step, the word count is calculated that provide a clear picture of the features of the dataset in terms of language.

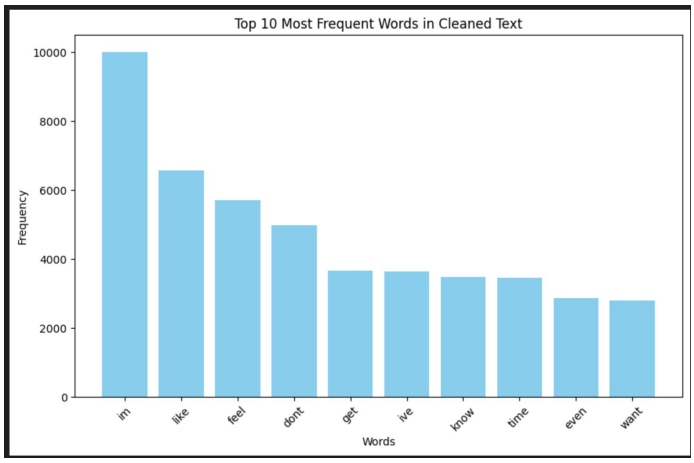


Fig. 14. Top frequent words after text preprocessing

Interaction of user with Streamlit: The user interface part is implemented in Streamlit, allows an interaction through text

and voice inputs. It can accommodate the real time analysis and visualization of responses and the related documents thus making the system light and easy to use.

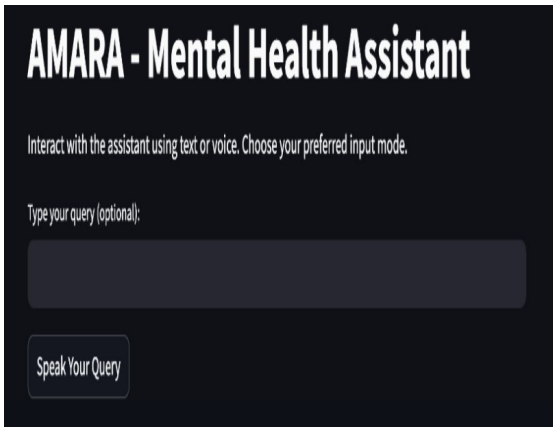


Fig. 15. UI of Streamlit

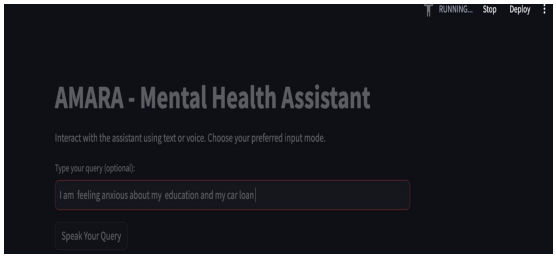


Fig. 16. Giving an Input query to Streamlit

Box Plot Analysis: The below image shows a box plot comparing text lengths across four categories: stress, personality disorder, depression, bipolar disease. Depression varies most and has the median length of the case, while bipolar disorder according to the diagram is the most consistent and compact with minimal median length.

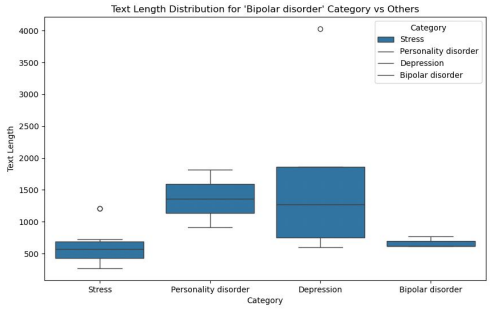


Fig. 17. Bipolar vs Other target features

Histogram Distribution: The image shows a simple histogram of the text lengths, on which density curves for various categories are plotted on. The distribution of text lengths is positively skewed, although it has a clear peak at texts with depths ranging between 500 and 1500. The extremes are observed at 500 characters and several values are below this limit and go up to 4000 characters.

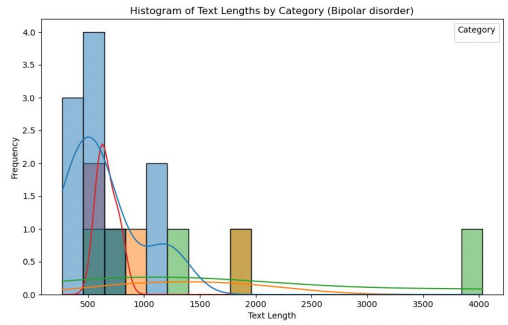


Fig. 18. Histogram Distribution

Violin Plot Visualization: The image accompanies violin plots of the probabilities of textual information density in terms of word count for all the categories. Distribution: It divides the shape of a curve to show that depression has more spread distribution than bipolar. The plots show how the spread and the center of mass differ between the categories.

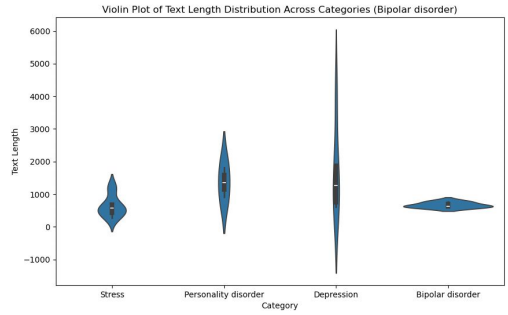


Fig. 19. Violin plot disribution

Statistical Significance: The image is Heatmap of pairwise t-test for p-value of the categories. Statistically significant (p¡0.05) between groups are shown in dark blue while non-significant difference is shown in red. Stress and bipolar disorder as the two categories that yield the highest p-value (0.88) implying that the difference between them is not high.

The stress category is tested by one way ANOVA test showing the F-statistic of 3.8137 and p-value of 0.8666. Thus, no differences are observed concerning text lengths between stress and other categories; therefore, the null hypothesis is not rejected. Pairwise t-test results demonstrate comparisons with personality disorder (t= 2.888; p= 0.0292), depression (t= 2.780; p= 0.0336) and bipolar disorder (t= 0.150; p= 0.8770). The second section analyzes the personality disorder category with F-statistic of 0.4457 and p-value of 0.5128. As with the stress category, there are no gross differences in text length with that of the personality disorder category. The pairwise comparisons show relatively similar t-test values and their connection with depression (t=1.7645) and bipolar disorder (t= 2.2448)

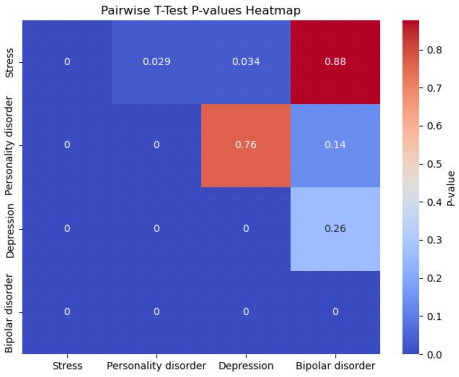
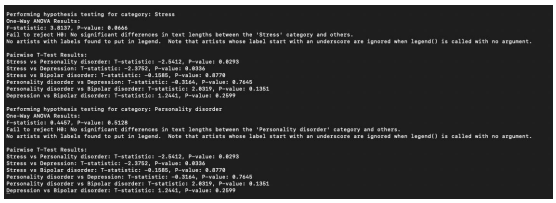


Fig. 20. Correlation in between the target features



```
Performing hypothesis testing for category: Stress
One-Way ANOVA Results:
F-statistic: 8.0197, P-value: 0.0666
Fail to reject H0: No significant differences in text lengths between the 'Stress' category and others.
No artists with labels found to put in legend.  Note that artists whose label start with an underscore are ignored when legend() is called with no argument.

Pairwise T-Test Results:
Stress vs Personality disorder: T-statistic: -2.5412, P-value: 0.0293
Stress vs Depression: T-statistic: -2.3752, P-value: 0.0336
Stress vs Bipolar disorder: T-statistic: -0.1585, P-value: 0.8770
Personality disorder vs Depression: T-statistic: -0.3164, P-value: 0.7645
Personality disorder vs Bipolar disorder: T-statistic: 2.0319, P-value: 0.1361
Depression vs Bipolar disorder: T-statistic: 1.2441, P-value: 0.2599

Performing hypothesis testing for category: Personality disorder
One-Way ANOVA Results:
F-statistic: 8.4487, P-value: 0.5128
Fail to reject H0: No significant differences in text lengths between the 'Personality disorder' category and others.
No artists with labels found to put in legend.  Note that artists whose label start with an underscore are ignored when legend() is called with no argument.

Pairwise T-Test Results:
Stress vs Personality disorder: T-statistic: -2.5412, P-value: 0.0293
Stress vs Depression: T-statistic: -2.3752, P-value: 0.0336
Stress vs Bipolar disorder: T-statistic: -0.1585, P-value: 0.8770
Personality disorder vs Depression: T-statistic: -0.3164, P-value: 0.7645
Personality disorder vs Bipolar disorder: T-statistic: 2.0319, P-value: 0.1361
Depression vs Bipolar disorder: T-statistic: 1.2441, P-value: 0.2599
```

Fig. 21. Pairwise T-test results

Depression category displays difference in text lengths with F=6.2241 and p=0.02251. It should be noted that the only hypothesis tested in this study that gave negative outcome was the null hypothesis which bespoke to statistically significant differences in the text length across the other categories. Bipolar Disorder Analysis: The category analysis of the bipolar disorder resulted to an 'F' statistic of 0.4717 and 'p' value of 0.50092. The null hypothesis is not to be rejected thus implying no considerable differences in text lengths between bipolar disorder and other categories.

```
Performing hypothesis testing for category: Depression
One-Way ANOVA Results:
F-statistic: 6.2241, P-value: 0.0225
Reject H0: Significant differences in text lengths between the 'Depression' category and others.
No artists with labels found to put in legend.  Note that artists whose label start with an underscore are ignored when legend() is called with no argument.

Pairwise T-Test Results:
Stress vs Personality disorder: T-statistic: -2.5412, P-value: 0.0293
Stress vs Depression: T-statistic: -2.3752, P-value: 0.0336
Stress vs Bipolar disorder: T-statistic: -0.1585, P-value: 0.8770
Personality disorder vs Depression: T-statistic: -0.3164, P-value: 0.7645
Personality disorder vs Bipolar disorder: T-statistic: 2.0319, P-value: 0.1361
Depression vs Bipolar disorder: T-statistic: 1.2441, P-value: 0.2599

Performing hypothesis testing for category: Bipolar disorder
One-Way ANOVA Results:
F-statistic: 0.4717, P-value: 0.5009
Fail to reject H0: No significant differences in text lengths between the 'Bipolar disorder' category and others.
No artists with labels found to put in legend.  Note that artists whose label start with an underscore are ignored when legend() is called with no argument.

Pairwise T-Test Results:
Stress vs Personality disorder: T-statistic: -2.5412, P-value: 0.0293
Stress vs Depression: T-statistic: -2.3752, P-value: 0.0336
Stress vs Bipolar disorder: T-statistic: -0.1585, P-value: 0.8770
Personality disorder vs Depression: T-statistic: -0.3164, P-value: 0.7645
Personality disorder vs Bipolar disorder: T-statistic: 2.0319, P-value: 0.1361
Depression vs Bipolar disorder: T-statistic: 1.2441, P-value: 0.2599
```

Fig. 22. Pairwise T-test results

These results suggest that stress has a PNTM text length that can be compared to a reference distribution but is significantly different from the distribution of personality disorder and depression, but not with bipolar disorder.

XI. Conclusion

AMARA Mental Health Assistant will leverage Natural Language Processing (NLP) to interpret mental health inquiries and provide tailored responses. The proposed approach utilizes Retrieval-Augmented Generation (RAG) and conversational memory to offer personalized advice and solutions drawn from an extensive medical text corpus. Key features include seamless integration of both textual and voice input, enhancing user accessibility, engagement, and overall mental health support.

Pairwise Comparisons

The most notable pairwise relationships include:

Comparison	T-statistic	P-value	Significance
Stress vs Personality disorder	-2.5412	0.0293	Significant
Stress vs Depression	-2.3752	0.0336	Significant
Stress vs Bipolar disorder	-0.1585	0.8770	Not Significant

Fig. 23. Pairwise comparison

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